

Open Single-Pixel, Event, and Image-Sensor Project Survey

HybridCamera Research Workspace

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Abstract

This survey identifies practical open-source and buyable building blocks for a hybrid scientific camera that combines a single-pixel bucket detector, event vision, and frame imaging. The central conclusion is pragmatic: single-pixel hardware is available in a credible open kit, event-camera software is open but event-sensor silicon is still commercial, and open frame-camera projects are the best source of PCB, enclosure, and synchronization patterns.

1 Single-Pixel Camera Projects

The strongest project found is ONE-PIX, an open single-pixel hyperspectral kit. It projects coded patterns onto a scene, records the reflected spectrum with a single detector/spectrometer path, and reconstructs a hyperspectral cube. Its software repository includes Raspberry Pi control, GUI and CLI workflows, examples, and plugin interfaces. Its hardware repository includes laser-cut box plans, STL fixtures, and editable CAD for adapting the instrument. This makes ONE-PIX the best reference for the HybridCamera bucket/spectral branch.

SPyRiT complements the hardware side. It is a Python toolbox for deep single-pixel image reconstruction, including neural reconstruction workflows that can be reused for coded bucket measurements. A small educational single-pixel-camera demo was also cloned as a minimal reference, but it should not be treated as the main instrument baseline.

2 Buying and Sourcing

For a complete single-pixel system, the best documented option is the premounted ONE-PIX kit linked from the hardware repository. UPOLabs also lists a DMD-based single-pixel imaging experiment system intended for teaching and research. For Taobao-style sourcing, the credible route is usually component sourcing rather than buying a polished single-pixel camera. The required parts are a DMD or DLP pattern source, photodiode/APD/PMT detector, low-noise transimpedance amplifier or lock-in amplifier, synchronized ADC/DAQ, lenses, calibration targets, and a rigid optical rail or enclosure. TI DLP LightCrafter controller boards are useful references for the DMD pattern-generation path.

3 Open Event-Camera Projects

No mature fully open event-sensor silicon project was found. The practical event path is to use commercial sensors or cameras with open software layers. OpenEB is the most important stack

because it exposes event-device abstraction, streaming, visualization, event manipulation, machine-learning utilities, and Prophesee camera plugins. The Prophesee/Sony IMX636 is a buyable event sensor family with evaluation kits and published product specifications.

jAER remains important for Address-Event Representation tooling and historical DVS/DAVIS workflows. v2e is useful before hardware arrives because it converts normal video into synthetic DVS events. E2VID is useful for reconstruction from events back to intensity frames. openEye-CamSI is the most relevant open hardware/gateway project because its roadmap explicitly targets an interface to an entry-level Prophesee/Sony event imaging sensor and Ethernet transfer.

4 Open Frame-Camera Hardware

Open frame-camera projects are more mature than open event-sensor hardware. The Antmicro OV9281 Dual Camera Board provides KiCad files for two global-shutter MIPI CSI-2 sensors. OneInchEye provides a Raspberry Pi camera board around the Sony IMX283 one-inch CMOS sensor, including KiCad and STEP assets. AXIOM Beta hardware provides a large modular open cinema-camera reference with sensor front-end boards, enclosure assets, and manufacturing files. Sseed’s open reCamera series is useful as a compact modular AI-camera hardware reference, but the cloned repository did not verify an event-camera board.

5 HybridCamera Implications

Branch	Use now	Later target
Single-pixel	ONE-PIX plus SPyRiT	Custom bucket detector and coded pattern API
Event	OpenEB-capable event camera	openEye-CamSI-style FPGA interface
Frame	OV9281, OneInchEye, reCamera, AXIOM references	Triggered global-shutter frame module
Fusion	Timestamped stream logs	Shared reconstruction objective across modalities

The first HybridCamera prototype should not wait for a custom ASIC. It should combine a central low-noise bucket detector, a commercial event camera or event module, and a frame camera under one hardware clock. Each record should include timestamp, source ID, exposure/gain, calibration profile, trigger metadata, and the optical pattern or scan position. Once this board-level data contract is stable, the project can decide whether a custom tile or chip is justified.

6 Shapr3D Export Guidance

For mechanical updates, export STEP as the main editable interchange format. Also export native Shapr3D files as the design archive and Parasolid files as a high-fidelity solid backup. Export DXF from sketches for 2D plates, optical mounts, laser-cut profiles, and PCB outline constraints. Export 3MF or STL only for printing and mesh preview. Export GLB or OBJ only for web previews, Blender renders, or documentation. IGES is a fallback if STEP fails. PDF, SVG, images, DWG, JT, and USDZ are useful only when a vendor or viewer specifically requests them.

The preferred package for this project is therefore: STEP, native Shapr3D, Parasolid, DXF sketches, and one 3MF or STL preview for each part.

7 Cloned Reference Set

- `external/one-pix`: single-pixel hyperspectral software.
- `external/one-pix-hardware`: ONE-PIX mechanical and build files.
- `external/spyrit`: single-pixel reconstruction toolbox.
- `external/openeb`: event-camera SDK modules.
- `external/event-based-get-started`: event-camera examples.
- `external/jaer`: AER/event-camera processing tools.
- `external/v2e`: video-to-event simulation.
- `external/rpg-e2vid`: event-to-frame reconstruction.
- `external/openeye-camsi`: open FPGA camera interface.
- `external/one-inch-eye`: open IMX283 frame-camera board.
- `external/ov9281-camera-board`: open OV9281 global-shutter board.
- `external/axiom-beta-hardware`: modular open camera hardware.
- `external/oshw-recamera-series`: modular open AI-camera hardware.

References

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- [6] Prophesee, “OpenEB,” <https://github.com/prophesee-ai/openeb>.
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- [10] UZH RPG, “E2VID,” https://github.com/uzh-rpg/rpg_e2vid.
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- [13] will127534, “OneInchEye,” <https://github.com/will127534/OneInchEye>.
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